

Projects Catalogue

Projects Catalogue Web Application
Enola Backend MariaDB Database Monitor User
Guide

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By David Egan

**Software
Version 1.0**

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1. Installation Enola

Installation of Enola is administrated from the Enola Installer Application. The installer handles both the client application and the service-based backend server installation automatically.

1.1 Prerequisites

- Windows 11 x64-bit
- MariaDB server instance accessible on the network
- Administrator privileges to run the installer
- projects catalogue web application for multi-user operation

1.2 Installing Enola Software

The Enola software comes with a built-in installer.

1. **Run the installer**
Right-click setup.exe and choose **Run as administrator**.
2. **Choose install location**
Keep the default path C:\Program Files\Enola\ and click **Next**.
3. **Set user access**
Select **Install for all users**, then click **Install**.
4. **Finish**
When installation completes, the Enola desktop icon will be created.

Enola is now installed. Double-click the desktop icon to launch the application.

1.3 Installation Logging

The date and time the service application is installed are logged in servicestatus.log. This file is generated and stored in the service subfolder within the EnolaApp directory under ProgramData:

C:\ProgramData\EnolaApp\service\servicestatus.log

2. Uninstalling Enola

Uninstalling Enola removes both the client application and the backend service automatically.

2.1 Uninstall Enola Software

1. **Launch Control Panel**
Press Win + R, type control, and press Enter.
2. **Open Programs and Features**
Click **Programs > Programs and Features**.
3. **Uninstall Enola**
Scroll to **Enola**, right-click, and choose **Uninstall**. Click **Yes** to confirm.

4. Done

Enola is now removed from your computer. The program files and desktop icon are also deleted automatically.

Important: If the Hidden Enola Services App is running as Primary Unlocker or the Enola client is running as Primary Unlocker on your desktop, the uninstaller will stop and remove them both automatically.

2.2 Uninstall Logging

The date and time the service application is removed are logged in servicestatus.log. This file is located in the service subfolder within the EnolaApp directory under ProgramData:

C:\ProgramData\EnolaApp\service\servicestatus.log

3. Overview

Enola Backend MariaDB Database Monitor is a Windows client application designed to automatically unlock records in the projects and user_accounts tables. The Enola Backend MariaDB Database Monitor application by design is an asynchronous application. The Enola Backend MariaDB Database Monitor communicates with the MariaDB server via TCP/IP and the MySQL protocol on port 3306. It operates in conjunction with the multi-user version of the projects catalogue web application.

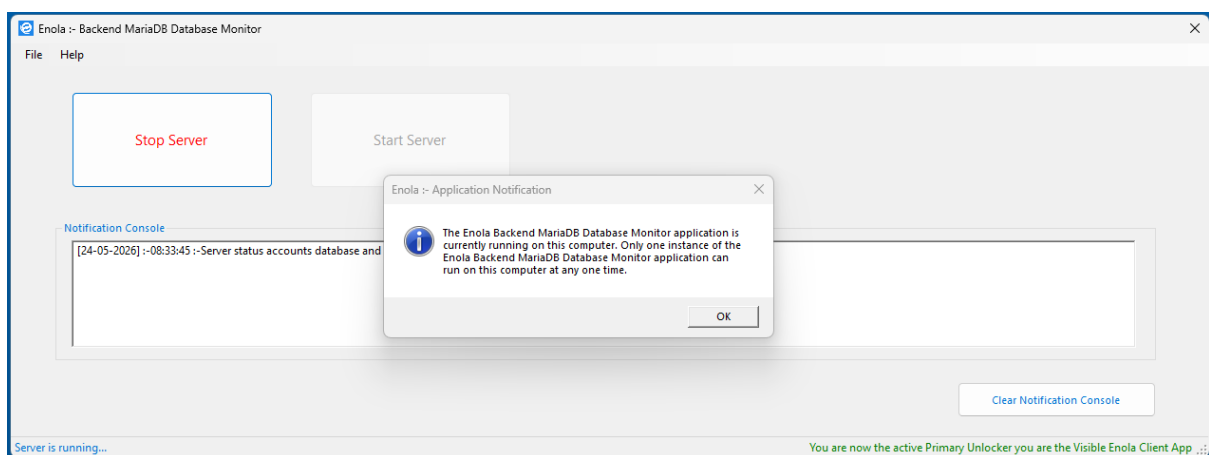


Figure 1

The dialog as shown in fig1 is displayed when you try and launch two or more instances of the Enola client application. Only one instance of the Enola Backend MariaDB Database Monitor client application can run on this computer at any one time.

4. Main Interface Components

The main window contains the following key areas:

Component	Description
Stop Server / Start Server Buttons	Toggle the backend monitoring service on or off. Only one button is active at a time.
Notification Console	Displays timestamped status messages from the server, including database and table checks.
Clear Notification Console	Clears all messages from the Notification Console window.
Status Bar - Left	Shows current server state: "Server is running..." or "Server is stopped..."
Status Bar - Right	Shows Primary Unlocker status. Green indicates the Enola client is active. Red indicates the Enola services app is active.
File Menu	Provides keyboard shortcuts for Start Server (Ctrl+E), Stop Server (Ctrl+A), and Exit (Ctrl+X).
Help Menu	Access the About dialog and additional system information.

5. Starting and Stopping the Server

5.1 Server Running State

When the server is running, the Stop Server button is active and shown in red. The Start Server button is disabled. The status bar will display "Server is running..." in blue text. The Notification Console logs operational messages.

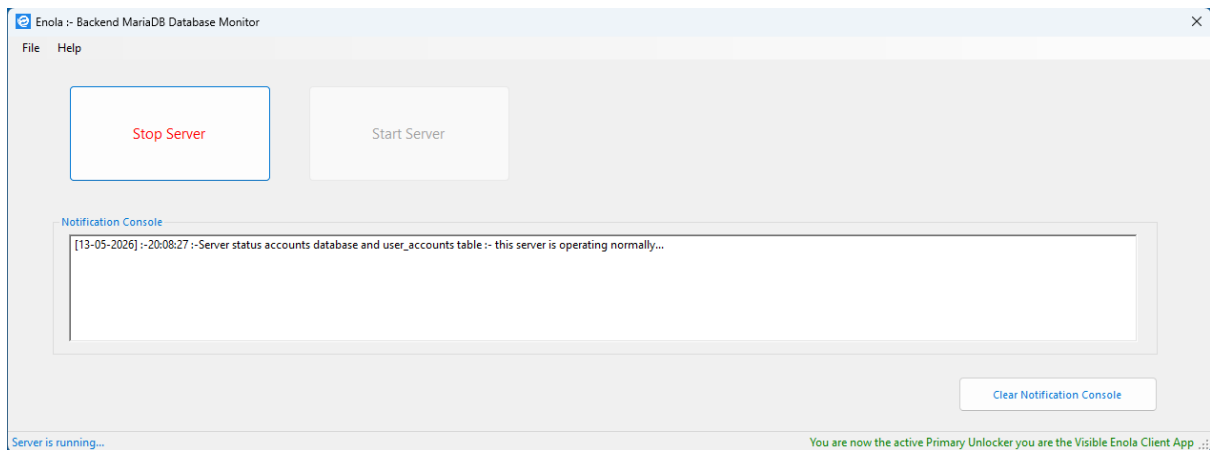


Figure 2: Enola Client with Server Running and Primary Unlocker Active

5.2 Server Stopped State

When the server is stopped, the Start Server button is active and shown in green. The Stop Server button is disabled. The status bar will display "Server is stopped..." in blue text. The Primary Unlocker role shifts to the Hidden Enola Services App.

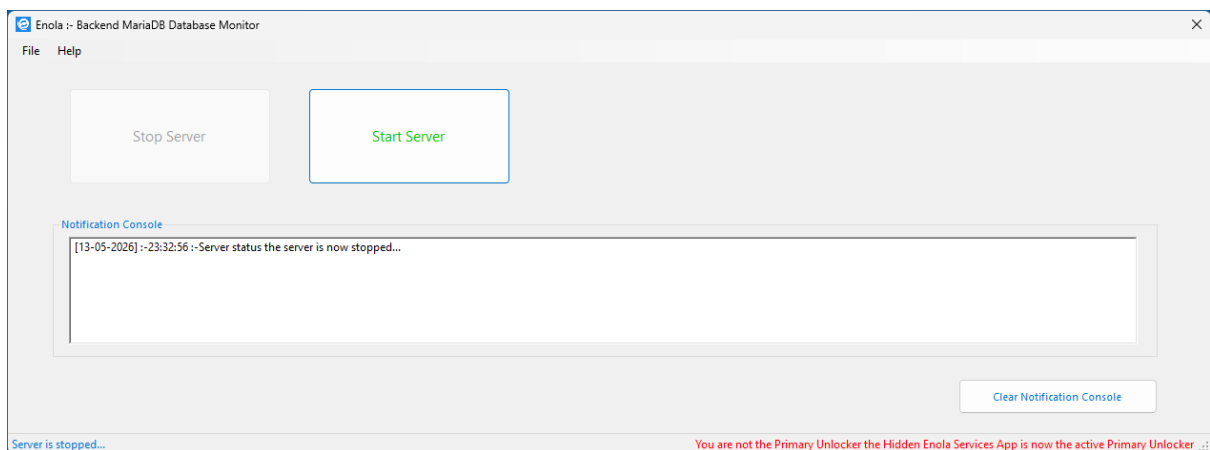


Figure 3: Enola Client with Server Stopped

6. Understanding Primary Unlocker Status

The Primary Unlocker is the service instance currently responsible for unlocking records in the projects or user_accounts tables. Only one Primary Unlocker can be active at a time.

Active Client: When you see green text "You are now the active Primary Unlocker you are the Visible Enola Client App", this client window controls the unlocking process as shown in fig 4 below.

Background Service Active: When you see red text "You are not the Primary Unlocker the Hidden Enola Services App is now the active Primary Unlocker", the background service has taken control. This typically occurs when the client server is stopped or is not active (visible on screen) on the user's desktop as shown in fig 5 below.

7. File Menu Shortcuts

For faster operation, use the File menu or keyboard shortcuts:

Ctrl+E - Start Server

Ctrl+A - Stop Server

Ctrl+X - Exit Application

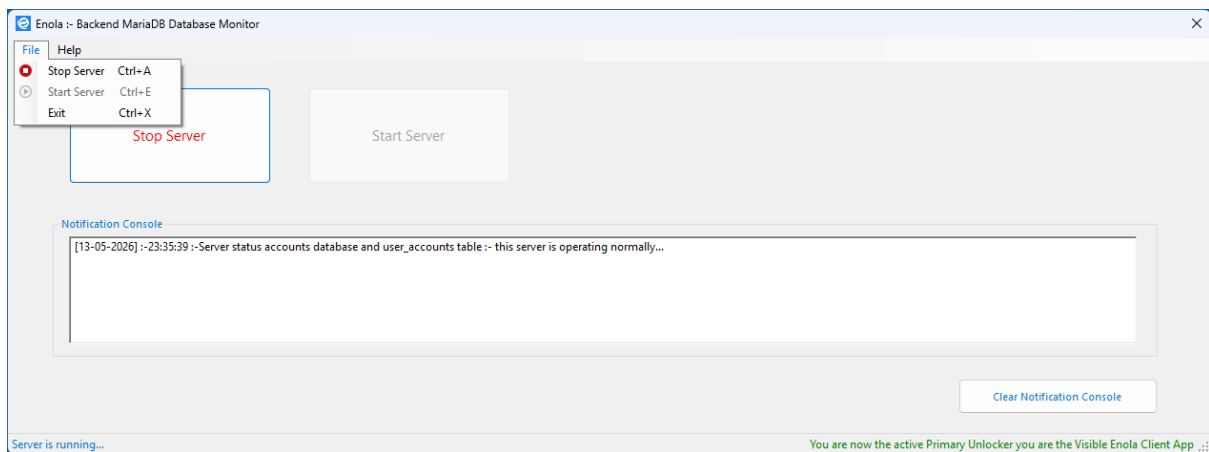


Figure 4: File Menu with Server Running

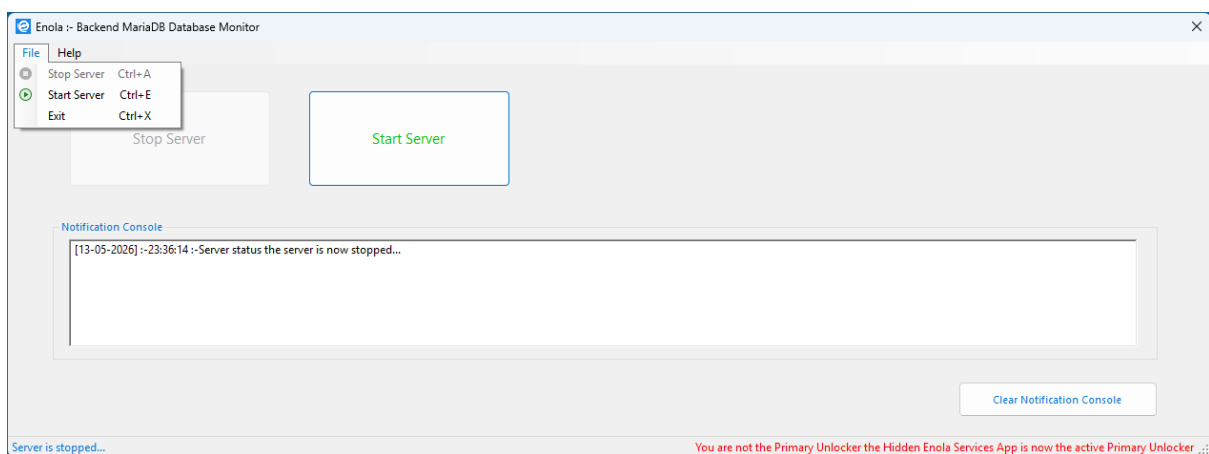


Figure 5: File Menu with Server Stopped

8. Using the Notification Console

The Notification Console provides a real-time log of server activity. Each entry is timestamped and includes status details for the projects database and projects table and accounts database and user_accounts table.

Common messages include:

- "Server status projects database and projects table -- this server is operating normally..."
- "Server status accounts database and user_accounts table -- this server is operating normally..."
- "Server status the server is now stopped..."

Use the "Clear Notification Console" button to clear the Notification Console window.

9. Help and About

Navigate to Help > About Enola to view application details, including version, platform, and registered owner.

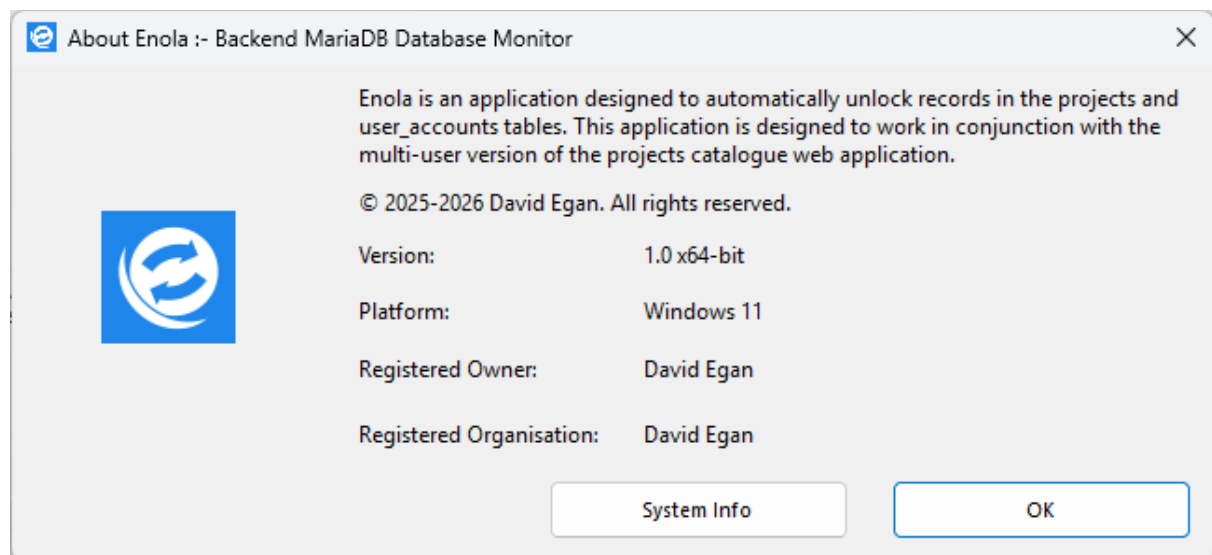


Figure 6: About Enola Dialog Box

10. Log Files and Diagnostics

Enola generates and stores log files in the service subfolder within the EnolaApp directory under ProgramData. The main service status log is located at:

C:\ProgramData\EnolaApp\service\servicestatus.log

This file records the date and time the service application is installed or removed. Review this log if you need to audit changes or diagnose issues.

10.1 Instance Logs

Enola generates and stores logs for the visible client and hidden service within the EnolaApp directory under ProgramData. These logs are located at:

C:\ProgramData\EnolaApp\logs\enola_instance_log_visible_pid_8324.log

C:\ProgramData\EnolaApp\logs\enola_instance_log_hidden_pid_5236.log

Enola generates separate instance logs for the visible client and hidden service. These are useful for diagnosing issues specific to each process:

- Visible Client App Log: enola_instance_log_visible_pid_8324.log
- Hidden Service App Log: enola_instance_log_hidden_pid_5236.log

These logs contain process-specific details such as PID, database queries, and unlock operations. Provide these logs to support when troubleshooting.

11. Troubleshooting

Issue	Resolution
Installation fails	Ensure you run the Enola Installer Application as Administrator and that MariaDB server is reachable. Check C:\ProgramData\EnolaApp\service\servicestatus.log for errors.
Uninstall fails or leaves files behind	Stop the server first, close the client, then run Uninstall Enola from Programs and Features Control Panel. Reboot if needed. Check servicestatus.log for removal events.
Start Server button is disabled	The server is already running. Click Stop Server first if you need to restart.
Status shows Hidden Enola Services App is active	This is expected when the client server is stopped. Start the server to make the Enola client the Primary Unlocker.
No messages in Notification Console	The server may be stopped. Check the status bar. Click Start Server to begin logging.
Application will not exit	Stop the server first using the Stop Server button or Ctrl+A, then use File > Exit or Ctrl+X.
Need to verify install/remove dates	Check C:\ProgramData\EnolaApp\service\servicestatus.log for timestamped install and uninstall entries.
Need detailed diagnostics	Review the instance logs: enola_instance_log_visible_pid_8324.log and enola_instance_log_hidden_pid_5236.log. Send both to support.

12. Log files

12.1 enola_instance_log_visible_pid_8324.log

```
[14-05-2026 05:48:23] PID 8324 [visible - enola] | Enola :- Backend MariaDB Database Monitor :- © 2025-2026 David Egan. All rights reserved. :-  
[14-05-2026 05:48:23] PID 8324 [visible - enola] | Visible Enola App :- Frontend Client Side :-  
[14-05-2026 05:48:23] PID 8324 [visible - enola] | Startup: StartHidden = False, IsPrimaryUnlocker = True  
[14-05-2026 05:50:10] PID 8324 [visible - enola] | [UNLOCKED] Project record unlocked for :- 20K156 - Milltown Wastewater Treatment Expansion » Lockedby :- David Egan - lockedon :- 14/05/2026 - at :- 05:45:09  
[14-05-2026 05:50:10] PID 8324 [visible - enola] | [NOTIFY] Project record unlocked for :- 20K156 - Milltown Wastewater Treatment Expansion » Lockedby :- David Egan - lockedon :- 14/05/2026 - at :- 05:45:09  
[14-05-2026 05:50:16] PID 8324 [visible - enola] | Server status :- the server is now stopped  
[14-05-2026 05:50:20] PID 8324 [visible - enola] | Server status :- the server has now started  
[14-05-2026 05:50:29] PID 8324 [visible - enola] | Server status :- the server is now stopped  
[14-05-2026 05:52:40] PID 8324 [visible - enola] | Visible Enola App has shutdown successfully on application exit
```

Note: The unlock time is 5 minutes, with a +2/1 second tolerance as reflected in the log above. This latency is caused by the display delay when the unlocked record appears in the Enola client notification console. Upon application exit, the log above indicates that the Enola client application shut down successfully. It safely disconnects from the databases and tables prior to shutting down. This behavior is critical to prevent database or table corruption.

Note: The visible log above is generated by the Enola client application and aligns with the 10-second demonstration video available on YouTube.

12.2 enola_instance_log_hidden_pid_5236.log

```
[14-05-2026 08:11:55] PID 5236 [hidden - enola] | Enola :- Backend MariaDB Database Monitor :- © 2025-2026 David Egan. All rights reserved. :-  
[14-05-2026 08:11:55] PID 5236 [hidden - enola] | Hidden Enola App :- Backend Server Service Side :-  
[14-05-2026 08:11:55] PID 5236 [hidden - enola] | Startup: StartHidden = true, IsPrimaryUnlocker = True  
[14-05-2026 08:11:55] PID 5236 [hidden - enola] | [UNLOCKED] Project record unlocked for :- 24K101 - Riverdale Bridge Structural Assessment » Lockedby :- David Egan - lockedon :- 14/05/2026 - at :- 08:06:55
```

Note: The unlock time is precisely 5 minutes, as indicated in the hidden log above.

12.3 servicestatus.log

```
[14-05-2026 14:37:25] PID 676 [service - status] | Enola :- Backend MariaDB Database Monitor :- © 2025-2026 David Egan. All rights reserved. :-  
[14-05-2026 14:37:25] PID 676 [service - status] | Hidden Enola App :- Backend Server Service Side :-  
[14-05-2026 14:37:25] PID 676 [service - status] | Startup: StartHidden = true, IsPrimaryUnlocker = true  
[14-05-2026 14:37:25] PID 676 [service - status] | [SUCCESS] The Enola Backend MariaDB Database Monitor service was removed successfully  
[14-05-2026 14:37:43] PID 168 [service - status] | [SUCCESS] The Enola Backend MariaDB Database Monitor service has been created and started successfully
```

Note: As shown in the servicestatus log above, the Enola service has been successfully removed, created, and started.

13. Log files Sign-out and Shutdown

13.1 enola_instance_log_visible_pid_4528.log sign-out

[14-05-2026 23:20:38] PID 4528 [visible - enola] | Enola :- Backend MariaDB Database Monitor :- © 2025-2026 David Egan. All rights reserved. :-
[14-05-2026 23:20:38] PID 4528 [visible - enola] | Visible Enola App :- Frontend Client Side :-
[14-05-2026 23:20:38] PID 4528 [visible - enola] | Startup: StartHidden = False, IsPrimaryUnlocker = True
[14-05-2026 23:20:52] PID 4528 [visible - enola] | Visible Enola App has shutdown successfully on PC Sign out

Note: When a sign-out is initiated while the Enola client application is the Primary Unlocker, it safely disconnects from the databases and tables prior to shutting down. This behavior is critical to prevent database or table corruption. The application establishes connections to the databases and tables at one-second intervals.

13.2 enola_instance_log_visible_pid_5140.log shutdown

[14-05-2026 23:42:36] PID 5140 [visible - enola] | Enola :- Backend MariaDB Database Monitor :- © 2025-2026 David Egan. All rights reserved. :-
[14-05-2026 23:42:36] PID 5140 [visible - enola] | Visible Enola App :- Frontend Client Side :-
[14-05-2026 23:42:36] PID 5140 [visible - enola] | Startup: StartHidden = False, IsPrimaryUnlocker = True
[14-05-2026 23:42:47] PID 5140 [visible - enola] | Visible Enola App has shutdown successfully on PC Powerdown

Note: When a shutdown is initiated while the Enola client application is the Primary Unlocker, it safely disconnects from the databases and tables prior to shutting down. This behavior is critical to prevent database or table corruption. The application connects to the databases and tables at one-second intervals.

13.3 enola_instance_log_hidden_pid_5260.log sign-out

[14-05-2026 23:22:00] PID 5260 [hidden - enola] | Enola :- Backend MariaDB Database Monitor :- © 2025-2026 David Egan. All rights reserved. :-
[14-05-2026 23:22:00] PID 5260 [hidden - enola] | Hidden Enola App :- Backend Server Service Side :-
[14-05-2026 23:22:00] PID 5260 [hidden - enola] | Startup: StartHidden = true, IsPrimaryUnlocker = True
[14-05-2026 23:22:00] PID 5260 [hidden - enola] | Hidden Enola App has shutdown successfully on PC Sign out

Note: When a sign-out is initiated while the Enola service application is the Primary Unlocker, it safely disconnects from the databases and tables prior to shutting down. This behavior is critical to prevent database or table corruption. The application establishes connections to the databases and tables at one-second intervals.

13.4 enola_instance_log_hidden_pid_5260.log shutdown

[14-05-2026 23:42:49] PID 5260 [hidden - enola] | Enola :- Backend MariaDB Database Monitor :- © 2025-2026 David Egan. All rights reserved. :-
[14-05-2026 23:42:49] PID 5260 [hidden - enola] | Hidden Enola App :- Backend Server Service Side :-
[14-05-2026 23:42:49] PID 5260 [hidden - enola] | Startup: StartHidden = true, IsPrimaryUnlocker = True
[14-05-2026 23:42:49] PID 5260 [hidden - enola] | Hidden Enola App has shutdown successfully on PC Sign out
[14-05-2026 23:42:55] PID 5260 [hidden - enola] | Hidden Enola App has shutdown successfully on PC Powerdown

Note: When a shutdown is initiated while the Enola service application is the Primary Unlocker, it safely disconnects from the databases and tables prior to shutting down. This behavior is critical to prevent database or table corruption. The application connects to the databases and tables at one-second intervals.

14. Log files Enola Client Errors

14.1 enola_instance_log_visible_pid_3980.log errors

```
[14-05-2026 05:11:52] PID 3980 [visible - enola] | Enola :- Backend MariaDB Database Monitor :- © 2025-2026 David Egan. All rights reserved. :-  
[14-05-2026 05:11:52] PID 3980 [visible - enola] | Visible Enola App :- Frontend Client Side :-  
[14-05-2026 05:11:52] PID 3980 [visible - enola] | Startup: StartHidden = False, IsPrimaryUnlocker = True  
[14-05-2026 05:13:25] PID 3980 [visible - enola] | Exception in ConnectProjects: Unknown database 'projects'  
[14-05-2026 05:13:28] PID 3980 [visible - enola] | Exception in ConnectProjects: Unknown database 'projects'  
[14-05-2026 05:13:31] PID 3980 [visible - enola] | Exception in ConnectProjects: Unknown database 'projects'  
[14-05-2026 05:13:34] PID 3980 [visible - enola] | Exception in ConnectProjects: Table 'projects.projects' doesn't exist  
[14-05-2026 05:13:46] PID 3980 [visible - enola] | Exception in ConnectProjects: Table 'projects.projects' doesn't exist  
[14-05-2026 05:13:49] PID 3980 [visible - enola] | Exception in ConnectProjects: Table 'projects.projects' doesn't exist  
[14-05-2026 05:13:52] PID 3980 [visible - enola] | Exception in ConnectProjects: Table 'projects.projects' doesn't exist  
[14-05-2026 05:13:55] PID 3980 [visible - enola] | Exception in ConnectProjects: Table 'projects.projects' doesn't exist  
[14-05-2026 05:14:58] PID 3980 [visible - enola] | Exception in ConnectProjects: Unable to connect to any of the specified MySQL hosts.  
[14-05-2026 05:15:01] PID 3980 [visible - enola] | Exception in ConnectAccounts: Unable to connect to any of the specified MySQL hosts.  
[14-05-2026 05:16:21] PID 3980 [visible - enola] | [UNLOCKED] Project record unlocked for :- 20K156 - Milltown Wastewater Treatment Expansion » Lockedby :- David Egan - lockedon :- 18/05/2026 - at :- 05:11:21  
[14-05-2026 05:16:21] PID 3980 [visible - enola] | [NOTIFY] Project record unlocked for :- 20K156 - Milltown Wastewater Treatment Expansion » Lockedby :- David Egan - lockedon :- 18/05/2026 - at :- 05:11:21  
[14-05-2026 05:16:43] PID 3980 [visible - enola] | [UNLOCKED] User account unlocked for :- Ryan Rodgers » Lockedby :- David Egan - lockedon :- 18/05/2026 - at :- 05:11:40  
[14-05-2026 05:16:43] PID 3980 [visible - enola] | [NOTIFY] User account unlocked for :- Ryan Rodgers » Lockedby :- David Egan - lockedon :- 18/05/2026 - at :- 05:11:40  
[14-05-2026 05:19:05] PID 3980 [visible - enola] | Exception in ConnectAccounts: Unknown database 'accounts'  
[14-05-2026 05:19:08] PID 3980 [visible - enola] | Exception in ConnectAccounts: Unknown database 'accounts'  
[14-05-2026 05:19:11] PID 3980 [visible - enola] | Exception in ConnectAccounts: Unknown database 'accounts'  
[14-05-2026 05:19:14] PID 3980 [visible - enola] | Exception in ConnectAccounts: Unknown database 'accounts'  
[14-05-2026 05:19:17] PID 3980 [visible - enola] | Exception in ConnectAccounts: Unknown database 'accounts'  
[14-05-2026 05:19:26] PID 3980 [visible - enola] | Exception in ConnectAccounts: Table 'accounts.user_accounts' doesn't exist  
[14-05-2026 05:19:29] PID 3980 [visible - enola] | Exception in ConnectAccounts: Table 'accounts.user_accounts' doesn't exist  
[14-05-2026 05:19:32] PID 3980 [visible - enola] | Exception in ConnectAccounts: Table 'accounts.user_accounts' doesn't exist  
[14-05-2026 05:19:54] PID 3980 [visible - enola] | Visible Enola App has shutdown successfully on application exit
```

Note: When the Enola client application operates as the Primary Unlocker, deletion of the projects database and projects table, or the accounts database and user_accounts table (as shown above), results in the exception displayed. Unable to connect to any of the specified MySQL hosts error indicates that the MySQL host is unreachable, likely because the MariaDB server has been shut down. Upon application exit, the log above indicates that the Enola client application shut down successfully. It safely disconnects from the databases and tables prior to shutting down. This behavior is critical to prevent database or table corruption.

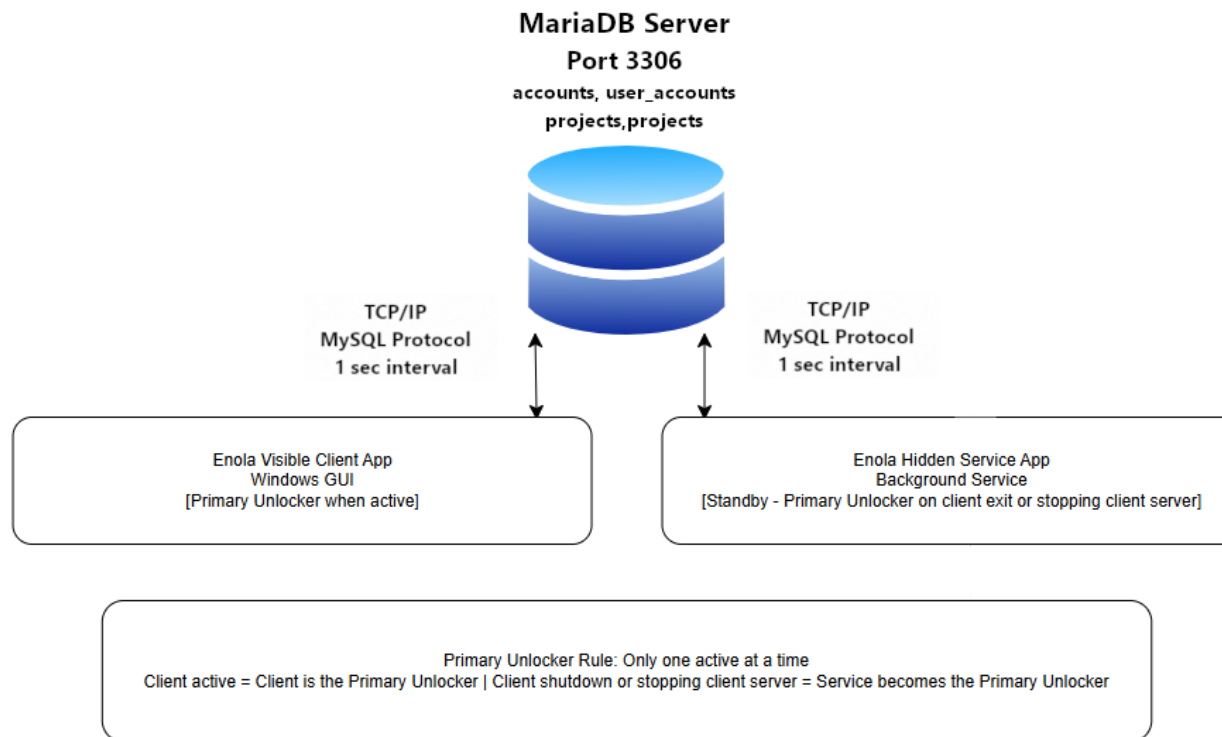
15. Log files Enola Service Errors

15.1 enola_instance_log_hidden_pid_5152.log errors

```
[14-05-2026 05:35:32] PID 5152 [hidden - enola] | Enola :- Backend MariaDB Database Monitor :- © 2025-2026 David Egan. All rights reserved. :-  
[14-05-2026 05:35:32] PID 5152 [hidden - enola] | Hidden Enola App :- Backend Server Service Side :-  
[14-05-2026 05:35:32] PID 5152 [hidden - enola] | Startup: StartHidden = true, IsPrimaryUnlocker = True  
[14-05-2026 05:35:32] PID 5152 [hidden - enola] | Exception in ConnectProjects: Unknown database 'projects'  
[14-05-2026 05:35:33] PID 5152 [hidden - enola] | Exception in ConnectProjects: Unknown database 'projects'  
[14-05-2026 05:35:34] PID 5152 [hidden - enola] | Exception in ConnectProjects: Unknown database 'projects'  
[14-05-2026 05:35:35] PID 5152 [hidden - enola] | Exception in ConnectProjects: Unknown database 'projects'  
[18-05-2026 05:35:39] PID 5152 [hidden - enola] | Exception in ConnectProjects: Unknown database 'projects'  
[14-05-2026 05:35:40] PID 5152 [hidden - enola] | Exception in ConnectProjects: Unknown database 'projects'  
[14-05-2026 05:35:41] PID 5152 [hidden - enola] | Exception in ConnectProjects: Unknown database 'projects'  
[14-05-2026 05:35:51] PID 5152 [hidden - enola] | Exception in ConnectProjects: Table 'projects.projects' doesn't exist  
[14-05-2026 05:35:52] PID 5152 [hidden - enola] | Exception in ConnectProjects: Table 'projects.projects' doesn't exist  
[14-05-2026 05:35:53] PID 5152 [hidden - enola] | Exception in ConnectProjects: Table 'projects.projects' doesn't exist  
[14-05-2026 05:35:54] PID 5152 [hidden - enola] | Exception in ConnectProjects: Table 'projects.projects' doesn't exist  
[14-05-2026 05:35:55] PID 5152 [hidden - enola] | Exception in ConnectProjects: Table 'projects.projects' doesn't exist  
[14-05-2026 05:35:56] PID 5152 [hidden - enola] | Exception in ConnectProjects: Table 'projects.projects' doesn't exist  
[14-05-2026 05:35:57] PID 5152 [hidden - enola] | Exception in ConnectProjects: Table 'projects.projects' doesn't exist  
[14-05-2026 05:35:58] PID 5152 [hidden - enola] | Exception in ConnectProjects: Table 'projects.projects' doesn't exist  
[14-05-2026 05:36:50] PID 5152 [hidden - enola] | Exception in ConnectProjects: Unable to connect to any of the specified MySQL hosts.  
[14-05-2026 05:36:52] PID 5152 [hidden - enola] | Exception in ConnectAccounts: Unable to connect to any of the specified MySQL hosts.  
[14-05-2026 05:39:59] PID 5152 [hidden - enola] | [UNLOCKED] Project record unlocked for :- 24K101 - Riverdale Bridge Structural Assessment » Lockedby :- David Egan - lockedon :- 18/05/2026 - at :- 05:34:59  
[14-05-2026 05:48:08] PID 5152 [hidden - enola] | Exception in ConnectAccounts: Table 'accounts.user_accounts' doesn't exist  
[14-05-2026 05:48:09] PID 5152 [hidden - enola] | Exception in ConnectAccounts: Table 'accounts.user_accounts' doesn't exist  
[14-05-2026 05:48:10] PID 5152 [hidden - enola] | Exception in ConnectAccounts: Table 'accounts.user_accounts' doesn't exist  
[14-05-2026 05:48:17] PID 5152 [hidden - enola] | Exception in ConnectAccounts: Table 'accounts.user_accounts' doesn't exist  
[14-05-2026 05:48:53] PID 5152 [hidden - enola] | Exception in ConnectAccounts: Unknown database 'accounts'  
[14-05-2026 05:48:54] PID 5152 [hidden - enola] | Exception in ConnectAccounts: Unknown database 'accounts'  
[14-05-2026 05:48:55] PID 5152 [hidden - enola] | Exception in ConnectAccounts: Unknown database 'accounts'
```

Note: When the Enola service application operates as the Primary Unlocker, deletion of the projects database and projects table, or the accounts database and user_accounts table (as shown above), results in the exception displayed. Unable to connect to any of the specified MySQL hosts error indicates that the MySQL host is unreachable, likely because the MariaDB server has been shut down.

16. Architecture Diagram - Primary Unlocker Model



Key Behaviors

1. **Primary Unlocker:** Only one instance can unlock records in the **projects** and **user_accounts** tables at a time.
2. **Client Active:** Enola Visible Client App holds Primary Unlocker status. Status bar shows **Green**.
3. **Client Shutdown or stopping client server:** Enola Hidden Service App becomes Primary Unlocker status. Status bar shows **Red**.
4. **Connection:** Both Apps connect to the MariaDB Server via TCP/IP on port 3306 at 1-second intervals.

